Stochastic Signal Analysis

Chapter 2 Review Problem Set

Eight-paper edition – 27 problems – 9 days – 3 problems/day

This set is based on eight verified past papers (file codes 201812, 201912, 202104, 202112, 202212, 202312, 202412, 202512). Chapter 2 carries 15.625 primary marks per paper in this sample; the recent-four mean is 9.00. All current homework questions are included. Added older-paper questions fill topic gaps. Adaptations explicitly state extra assumptions and extensions. Complete one page at a time, leaving the blank space for your derivation.

2026-10-13

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Source tags distinguish verbatim homework, original exam excerpts, adaptations, and recall exercises. Day labels are a study schedule, not the order of the original exams.

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Problem 1 — Day 1

(Recall / extension)

Textbook sections: 2.4.1–2.4.3

Define a Gaussian process using its finite-dimensional distributions. For a Gaussian vector with mean $\boldsymbol{\mu}$ and covariance C, state its characteristic function, its ordinary PDF (including the condition for its existence), and the law of $G\boldsymbol{X} + \boldsymbol{b}$.
Is marginal Gaussianity sufficient for joint Gaussianity? Give a counterexample. State when uncorrelatedness implies independence.

Working space

Problem 2 — Day 1

(Textbook 2.19 / Required)

Textbook sections: 2.4.1

Let the random vector $(X, Y) \sim N(\boldsymbol{\mu}, \boldsymbol{C})$, where

$$\boldsymbol{\mu} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \qquad \boldsymbol{C} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}.$$

Find the joint probability density function of (X, Y) and its characteristic function $\phi_{XY}(u, v)$.

Working space

Problem 3 — Day 1

(Textbook 2.17 / translated)

Textbook sections: 2.4.1 / 2.4.3

Let $X(t)$ be a zero-mean Gaussian signal and put $X_k = X(t_k)$ for $k = 1, 2, 3, 4$. Prove that

$$E[X_1X_2X_3X_4] = E[X_1X_2]E[X_3X_4] + E[X_1X_3]E[X_2X_4] + E[X_1X_4]E[X_2X_3].$$

Working space

Problem 4 — Day 2

(Adapted from 2022–23 Q3)

Textbook sections: 2.4.3

A zero-mean stationary Gaussian white noise $N(t)$ has PSD $N_0/2$, where $N_0 > 0$. It passes through an RC low-pass filter

$$H(j\omega) = \frac{1}{1 + j\omega RC}, \qquad R, C > 0,$$

whose output is $X(t)$. A second device produces $Y(t) = aX^2(t)$, where a is real.

1. Find $R_X(\tau)$.
2. Find $E[Y(t)]$, $R_Y(\tau)$ and $C_Y(\tau)$.
3. Is $Y(t)$ Gaussian when $a \neq 0$?

The transfer function is supplied in place of the original circuit figure; the covariance and Gaussianity questions are extensions.

Working space

Problem 5 — Day 2

(Textbook 2.16 / Required)

Textbook sections: 2.4.1 / 2.4.3

Let the random process be

$$X(t) = A\cos\omega t + B\sin\omega t,$$

where ω is a constant and A and B are independent Gaussian random variables. Given that

$$E[A] = E[B] = 0, \qquad E[A^2] = E[B^2] = \sigma^2,$$

find the one-dimensional and two-dimensional probability density functions of $X(t)$.

Working space

Problem 6 — Day 2

(Textbook 2.18 / Optional)

Textbook sections: 2.4.1 / 2.4.3

A Gaussian signal has mean and covariance function

$$m_X(t) = 2, \qquad C_X(t_1, t_2) = 8\cos(t_1 - t_2).$$

Write down its three-dimensional probability density function at the times

$$t_1 = 0, \qquad t_2 = 0.5, \qquad t_3 = 1.$$

Working space

Problem 7 — Day 3

(Adapted from 2022–23 Q2(a,b))

Textbook sections: 2.2.1 / 2.5

Let $\eta(t) = \xi(t)\cos(\omega_0 t + \Theta)$, where $\xi(t)$ is a real WSS process with finite second moment and $\Theta \sim U[0, 2\pi]$ is independent of the entire process. Find the mean and autocorrelation of $\eta(t)$, and determine whether it is WSS. Show the phase averages explicitly. No PSD calculation is required.

Working space

Problem 8 — Day 3

(Adapted from 2024–25 Q1)

Textbook sections: 2.3.2 / 2.5

Let

$$X[n] = \sin(2\pi n + \Phi), \qquad Y[n] = \cos(2\pi n + \Phi), \qquad n \in \mathbb{Z},$$

where $\Phi \sim U(0, \pi)$.

1. Find $R_{XY}(m, n)$ and the two means.
2. Are the signals orthogonal, uncorrelated, or statistically independent? Justify each answer.
3. Are they jointly stationary?

Working space

Problem 9 — Day 3

(Textbook 2.11 / Optional)

Textbook sections: 2.3.2 / 2.5

Consider the two random signals

$$X(t) = A\sin(\omega t + \Theta), \qquad Y(t) = B\cos(\omega t + \Theta),$$

where A and B are random variables with unspecified distributions, Θ is uniformly distributed on $[0, 2\pi]$, and ω is a constant. The random variables A, B, and Θ are pairwise statistically independent.

1. Find the cross-correlation function $R_{XY}(t_1, t_2)$ of the two random signals.
2. Discuss their orthogonality, uncorrelatedness, and statistical independence.

Working space

Problem 10 Day 4

(Recall / extension)

Textbook sections: 2.1.3 / 2.3.3

For a real process $X(t)$ with finite second moments:

1. Define m_X, R_X and C_X, and relate $R_X(t,t)$ to variance.
2. Prove that every sampled covariance matrix is positive semidefinite, and that $|C_X(s,t)|^2 \leq C_X(s,s)C_X(t,t)$.
3. For $Y(t) = X(t+a) - X(t)$, expand $R_Y(s,t)$ without assuming stationarity.

Working space

Problem 11 — Day 4

(Adapted from 2022–23 Q1)

Textbook sections: 2.1.1–2.1.3

Let $X(t) = U + t$ for $-2 \leq t \leq 2$, where $U \sim U[0, \pi]$. Sketch two sample functions, find the characteristic function of $X(t)$, and compute its mean and autocovariance. Explain whether samples at distinct times are independent.

Working space

Problem 12 Day 4

(Recall / extension)

Textbook sections: 2.3.1 / 2.5

State the finite-dimensional definition of independence of two random signals. Does independence of $X(t)$ and $Y(t)$ at every equal time imply independence of the two signals? Use an iid sequence $\{Z_n\}$ of fair signs and $X_n = Z_n$, $Y_n = Z_{n-1}$ to answer. Also state the criterion for two jointly Gaussian signals to be independent.

Working space

Problem 13 — Day 5

(Textbook 2.3 / Required)

Textbook sections: 2.1.2 / 2.3.1

A coin is tossed, and a random process is defined by

$$X(t) = \begin{cases} \cos \pi t, & \text{if heads occurs,} \\ 2t, & \text{if tails occurs.} \end{cases}$$

Heads and tails are equally likely. Find:

1. The one-dimensional distribution functions $F_X(x, 1/2)$ and $F_X(x, 1)$ of $X(t)$;
2. The two-dimensional distribution function $F_X(x_1, x_2; 1/2, 1)$ of $X(t)$;
3. Sketch the graphs of the above distribution functions.

Working space

Problem 14 — Day 5

(Adapted from 2025–26 Q1)

Textbook sections: 2.1.2 / 2.3.1 / 2.5

A fair coin is tossed once. Define $X(t) = \sin(2\pi t)$ for heads and $X(t) = t^2$ for tails.

1. Find the generalized PDF of $X(2)$.
2. Find the joint generalized PDF of $(X(1), X(2))$.
3. Find $C_X(s,t)$.
4. Now define $Y[n] = \sin(2\pi n)$ or n^2 by a fresh independent fair toss at each integer n. Find the joint generalized PDF of $(Y[1], Y[2])$.

Working space

Problem 15 — Day 5

(Textbook 2.1 / Optional)

Textbook sections: 2.1.1 / 2.2.1

Consider the sinusoidal random signal $X(t) = V\cos\omega t$, where $0 \le t < \infty$, ω is a constant, and V is uniformly distributed on $[0, 1]$.

1. Sketch two sample functions of this process.
2. Determine the probability density function of the random variable $X(t_i)$ at each of the times
$$t_i = 0, \quad \frac{\pi}{4\omega}, \quad \frac{3\pi}{4\omega}, \quad \frac{\pi}{\omega}.$$
3. Find the probability density function of $X(t_0)$ when $t_0 = \pi/(2\omega)$.

Working space

Problem 16 — Day 6

(Textbook 2.4 / Optional)

Textbook sections: 2.2.2

Suppose that the binary data sequence $\{B(n), n = 1, 2, 3, \ldots\}$ is a Bernoulli random sequence. Each bit corresponds to a random variable $B(n)$, with

$$P[B(n) = 0] = 0.2, \qquad P[B(n) = 1] = 0.8.$$

1. What is the probability that four consecutive bits form the word $\{1011\}$?
2. What is the mean word formed by four consecutive bits?
3. Which word formed by four consecutive bits has the highest probability?

Working space

Problem 17 — Day 6

(Textbook 2.8 / Optional)

Textbook sections: 2.2.2 / 2.1.3

Suppose that the Bernoulli sequence $\{I_n, n = 1, 2, \ldots\}$ takes the values -1 and $+1$ with equal probabilities.
Find its one-dimensional probability density function, mean, and covariance function.

Working space

Problem 18 — Day 6

(Textbook 2.7 / Optional)

Textbook sections: 2.4.2 / 2.1.3

The position of a moving particle is described by the straight-line process

$$X(t) = Vt + X_0,$$

where $V \sim N(1, 1)$ and $X_0 \sim N(0, 2)$ are independent.
Find the one-dimensional probability density function, mean, and variance of the random variable $X(t)$ at time t.

Working space

Problem 19 — Day 7

(Textbook 2.13 / Required)

Textbook sections: 2.2.1 / 2.3.4

Consider the sinusoidal voltage signal

$$X(t) = A\cos(\omega t + \Theta),$$

where ω is a constant, $A \sim U(-1, +1)$, and $\Theta \sim U(-\pi, +\pi)$. The random variables A and Θ are independent.
If this voltage is applied across a parallel RC circuit, find the total current signal and its mean-square value.

Working space

Problem 20 — Day 7

(Textbook 2.14 / translated)

Textbook sections: 2.3.4

For the integral of a random signal,

$$Y(t) = \int_0^t X(u)\,du, \qquad t \geq 0,$$

find expressions for its mean and autocorrelation function. State sufficient conditions that justify interchanging integration and expectation.

Working space

Problem 21 — Day 7

(Adapted from 2023–24 Q2(c))

Textbook sections: 2.4.3

A zero-mean white noise $W(t)$ with PSD $N_0/2$ passes through $h(t) = e^{-t}u(t)$ to produce $X(t)$. Let $Y(t) = X(t) - X(t-T)$, with $T > 0$ and $N_0 > 0$. You may use $R_X(\tau) = N_0 e^{-|\tau|}/4$.

1. Do the stated assumptions imply that $Y(t)$ is Gaussian?
2. Under the additional assumption that W is Gaussian white noise, find the one-dimensional PDF of $Y(t)$.
3. Describe the case $T = 0$.

The extra Gaussian assumption in (2) is explicit; it was not stated in the original exam stem.

Working space

Problem 22 — Day 8

(Adapted from 2018–19 Q3 (201812))

Textbook sections: 2.2.1 / 2.4.2 / 2.4.3

Let $X(t) = A\cos(\omega t) + B\sin(\omega t)$, with independent $A, B \sim N(0, \sigma^2)$, $\sigma^2 > 0$ and $\omega \neq 0$.

1. Find $m_X(t)$ and $R_X(s,t)$.
2. Prove that X is Gaussian and WSS (Chapter 3 extension).
3. Are the component processes $A\cos(\omega t)$ and $B\sin(\omega t)$ individually WSS?

The original paper calls the first component stationary. This adaptation asks you to check that claim rather than assume it.

Working space

Problem 23 — Day 8

(2020–21 / 2021–22 Q1; extended (202104/202112))

Textbook sections: 2.4.1 / 2.4.3 / 2.1.3

Let $X(t) = A + B\cos t$, where A, B are independent $N(0,1)$ variables.

1. Find its mean, autocorrelation and one-dimensional PDF.
2. For fixed s, t, find the covariance matrix and joint PDF of $(X(s), X(t))$ whenever an ordinary PDF exists. Describe the singular case.
3. Evaluate the matrix and joint CF at $s = 0, t = \pi/2$. Is the process WSS?

Singular times and WSS are explicit extensions of the exam questions.

Working space

Problem 24 — Day 8

(Adapted from 2019–20 Q3 (201912))

Textbook sections: 2.2.3 / 2.1.3

Let B_k, $k \in \mathbb{Z}$, be independent Bernoulli variables with $P(B_k = 0) = P(B_k = 1) = 1/2$. For $T_0 > 0$, define the fixed-clock binary signal

$$X(t) = B_{\lfloor t/T_0 \rfloor}.$$

1. Find $m_X(t)$ and $R_X(t, t+\tau)$.
2. Is X WSS? Show the role of symbol boundaries.
3. As an extension, let $D \sim U(0, T_0)$ be independent and set $Y(t) = X(t - D)$. Find $R_Y(\tau)$.

The fixed-clock convention is explicit; randomizing the clock is a separate extension.

Working space

Problem 25 — Day 9

(Adapted from 2018–19 Q4 / 2019–20 Q4)

Textbook sections: 2.2.1 / 2.5; extension 4.1

Let $X(t) = A\cos(\omega t + \Theta)$, $\omega \neq 0$, with $\Theta \sim U(0, 2\pi)$, independent of a real random amplitude A. The amplitude is fixed during each realization and $E(A^2) < \infty$.

1. Find its mean and autocorrelation.
2. Find the time mean and time autocorrelation of a realization.
3. Under what condition is it correlation ergodic? Does mean ergodicity require the same condition?

Independence, constant-in-time amplitude, and finite second moment are explicit assumptions; the original allowed several amplitude models.

Working space

Problem 26 — Day 9

(Adapted from 2019–20 Q5 (201912))

Textbook sections: 2.2.1 / 2.4.3; extension 4.1

Let $X(t) = \operatorname{sgn}(A)\cos(\omega_0 t + \Theta)$, with independent $A \sim N(0,1)$ and $\Theta \sim U(0, 2\pi)$, $\omega_0 \neq 0$.

1. Find $m_X(t)$ and $R_X(s,t)$.
2. Is $X(t)$ Gaussian? Derive its one-dimensional PDF.
3. Compare its time and ensemble mean and autocorrelation (Chapter 4 extension).

The PDF and Gaussianity questions are extensions.

Working space

Problem 27 — Day 9

(2021–22 Q2(a,b); extensions (202112))

Textbook sections: 2.3.2 / 2.5; extension 3.1

Let X and N be independent real zero-mean WSS processes with finite second moments. A radar receiver observes

$$Y(t) = \alpha X(t-d) + N(t), \qquad d > 0, \quad \alpha \in \mathbb{R}.$$

1. Find $m_Y(t)$ and $R_Y(\tau)$.
2. With $R_{XY}(\tau) = E[X(t)Y(t+\tau)]$, find $R_{XY}(\tau)$ and $R_{YX}(\tau)$.
3. Are Y WSS and X, Y jointly WSS? Is Y necessarily Gaussian?

The time delay is renamed d to distinguish it from the correlation lag; Gaussianity is an extension.

Working space